

CM_2

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```
package CM2;

public class Pembeli {
    String namaPembeli;
    String noHp;
    int noPesanan = 0;
    Pesanan pesanan;

    Pembeli(String namaPembeli, String noHp){
        this.namaPembeli = namaPembeli;
        this.noHp = noHp;
    }

    public void tampilPembeli(){
        System.out.println(noPesanan + "          "+namaPembeli+"          "+
noHp);
    }
}
```

```
package CM2;

public class Pesanan {
    int kodePesanan;
    String namaPesanan;
    int harga;
    String namaPembeli;

    Pesanan(int kodePesanan, String namaPesanan, int harga, String namaPembeli){
        this.kodePesanan = kodePesanan;
        this.namaPesanan = namaPesanan;
        this.harga = harga;
        this.namaPembeli = namaPembeli;
    }
}
```

```

        public void tampilPesanan(){
            System.out.println(kodePesanan + "          " + namaPesanan + "          " +
harga + "          " + namaPembeli);
        }

    }
}

```

```

package CM2;

public class IniNode {
    Pembeli data;
    IniNode prev;
    IniNode next;

    public IniNode(Pembeli data){
        this.data = data;
        this.prev = null;
        this.next = null;
    }
}

```

(Node milik pembeli)

```

package CM2;

public class IniNodeJuga {
    Pesanan data;
    IniNodeJuga prev;
    IniNodeJuga next;

    public IniNodeJuga(Pesanan data){
        this.data = data;
        this.prev = null;
        this.next = null;
    }
}

```

(Node milik pesanan)

```

package CM2;

```

```

public class IniDLL {
    IniNode head;
    IniNode tail;
    int nomorPesanan = 1;
    Pesanan pesanan;
    public IniDLL(){
        head = null;
        tail = null;
    }
    public boolean isEmpty(){
        return head == null;
    }

    public void add(Pembeli data){

        data.noPesanan = nomorPesanan;
        nomorPesanan++;

        IniNode newNode = new IniNode(data);

        if (isEmpty()) {
            head = tail = newNode;
        }else {
            tail.next =newNode;
            newNode.prev = tail;
            tail = newNode;
        }
        System.out.println("Antrian berhasil ditambahkan dengan nomor : " +
data.noPesanan);
    }
    public void print(){
        if (isEmpty()) {
            System.out.println("Linked List masih kosong.");
            return;
        } System.out.println( "No.Pesanan   |   Nama       |   No.Hp   ");

        IniNode current = head;
        while (current != null) {

            current.data.tampilPembeli();
            current = current.next;
        }
    }
}

```

```

    }
    public Pembeli remove(){

        if (isEmpty()) {
            System.out.println("Linked List masih kosong, tidak dapat dihapus!");
            return null;
        }
        Pembeli data = head.data;

        if (head == tail) {
            head = tail = null;
        }else {
            head = head.next;
            head.prev = null;
        }
        return data;
    }
}

```

(DLL milik pembeli)

```

package CM2;

public class IniDLLJuga {
    IniNodeJuga head;
    IniNodeJuga tail;

    public IniDLLJuga(){
        head = null;
        tail = null;
    }

    public boolean isEmpty(){
        return head == null;
    }

    public void addLast(Pesanan data){

        IniNodeJuga newNode = new IniNodeJuga(data);
    }
}

```

```

        if(isEmpty()){
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
    }

    public void sortByNamaPesanan(){

        if(head == null){
            return;
        }

        boolean swapped;

        do{
            swapped = false;

            IniNodeJuga current = head;

            while(current.next != null){

                String a = current.data.namaPesanan;
                String b = current.next.data.namaPesanan;

                if(a.compareToIgnoreCase(b) > 0){

                    Pesanan temp = current.data;
                    current.data = current.next.data;
                    current.next.data = temp;

                    swapped = true;
                }

                current = current.next;
            }

        } while(swapped);
    }

    public void laporanPesanan(){

```

```

        if(isEmpty()){
            System.out.println("Belum ada pesanan!");
            return;
        }

        sortByNamaPesanan();

        System.out.println("==== LAPORAN PESANAN =====");
        System.out.println(
            "Kode      |      Nama Menu      |      Harga      |      Nama
Pembeli"
        );

        IniNodeJuga current = head;

        int total = 0;

        while(current != null){

            current.data.tampilPesanan();

            total += current.data.harga;

            current = current.next;
        }

        System.out.println("=====");
        System.out.println("Total Pendapatan : " + total);
    }
}

```

(DLL milik pesanan)

```

package CM2;

import java.util.Scanner;

public class IniMain {

```

```

public static Pembeli inputPembeli(Scanner sc){
    System.out.print("Masukkan Nama : ");
    String namaPembeli = sc.nextLine();

    System.out.print("Masukkan No.Hp : ");
    String noHp = sc.nextLine();

    return new Pembeli(namaPembeli, noHp);
}
public static Pesanan inputPesanan(Scanner sc, String namaPembeli){
    System.out.print("Kode Pesanan : ");
    int kodePesanan = sc.nextInt();
    sc.nextLine();

    System.out.print("Nama Pesanan : ");
    String namaPesanan = sc.nextLine();

    System.out.print("Harga : ");
    int harga = sc.nextInt();
    sc.nextLine();

    return new Pesanan(kodePesanan, namaPesanan, harga, namaPembeli);
}
public static void main(String[] args) {
    IniDLLJuga Pesanan = new IniDLLJuga();
    IniDLL list = new IniDLL();
    Scanner sc = new Scanner(System.in);
    int pilihan;

    do{
        System.out.println("===== Sistem Antrian Royal Delish =====");
        System.out.println("1.          Tambah Antrian          ");
        System.out.println("2.          Cetak Antrian          ");
        System.out.println("3.          Hapus Antrian dan Pesan ");
        System.out.println("4.          Laporan Pesanan        ");
        System.out.println("0.          Keluar                  ");
        System.out.println("=====");
        System.out.print("Pilihan : ");
        pilihan = sc.nextInt();
        sc.nextLine();

        switch (pilihan) {

```

```

case 1:
    Pembeli pmb1Awal = inputPembeli(sc);
    list.add(pmb1Awal);
    break;

case 2:
    list.print();
    break;

case 3:
    Pembeli keluar =
        list.remove();

    if(keluar != null){

        System.out.println(
            "Pembeli dipanggil:"
        );

        keluar.tampilPembeli();

        System.out.println(
            "=== Input Pesanan ==="
        );

        Pesanan pesanan =
            inputPesanan(sc, keluar.namaPembeli);

        Pesanan.addLast(pesanan);

        System.out.println(
            "Pesanan berhasil disimpan!"
        );
    }

    break;
case 4:
    Pesanan.laporanPesanan();
    break;

case 0:
    System.out.println("Program selesai.");
    break;

```

```
        default:
            System.out.println("Menu Ga Valid!");
        }
    }while (pilihan != 0);
    sc.close();
}
}
```